CS102L Data Structures and Algorithms
Review - Tuples

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Contents

1 Review Exercises **2**
1.1 Spin the Wheel 2
1.1.1 Background 2
1.1.2 Challenge 2
1.1.3 Note 2
1.1.4 Sample interaction 2
1.1.5 Constraints 3
1.1.6 Testing 3
1.2 Who wants to be a multipillionaire? 3
1.2.1 Background 3
1.2.2 Example 4
1.2.3 Challenge 4
1.2.4 Note 5
1.2.5 Sample interaction 5
1.2.6 Sample Interaction Explaination 5
1.2.7 Constraints 5
1.2.8 Testing 5

Review Exercises

1.1 Spin the Wheel

1.1.1 Background

On a particular game show, a contestant would spin a giant wheel in front of her that has numbers from 0 to 99. Each contestant initially starts from a given position on her wheel, called a standing, and then, on her turn, advances the wheel by a set number, called a jump. When a player's wheel lands on 99, the goal, the game ends, and the player who reaches the goal first is declared the winner.

For example, if Carol starts from an initial standing of 38, and advances by a jump of 23, then her spins will proceed as follows: 38, 61, 84, 7, 30, 53, 76, 99.

If Diana starts from an initial standing of 38, and advances by a jump of 87, then her spins will proceed as follows: 38, 25, 12, 99. Since Diana reaches her goal in 3 spins, she wins.

1.1.2 Challenge

Write a function named `spin_the_wheel` that accepts two arguments: standings (a tuple of ints – starting positions on the wheels for each player), and jumps (a tuple of ints – number of positions that the wheel advances by on each player's turn). The function computes the standings for each round and returns all standings as a list of tuples. When a player reaches the goal, the function prints a message declaring that player the winner of the game.

1.1.3 Note

It is entirely possible for certain values for initial standings and jumps to never reach the goal. It is also possible for two players to reach the goal in the same turn. In this case, the player with the initiative (player with the smaller position number) wins.

1.1.4 Sample interaction

```
>>> spin_the_wheel((99, 0), (7, 7))
Player 0 wins.
[(99, 0)]

>>> spin_the_wheel((0, 1), (7, 7))
Player 1 wins.
[(0, 1), (7, 8), (14, 15), (21, 22), (28, 29), (35, 36),
   (42, 43), (49, 50), (56, 57), (63, 64), (70, 71), (77,
   78), (84, 85), (91, 92), (98, 99)]

>>> spin_the_wheel((30, 47, 39), (54, 66, 6))
Player 2 wins.
[(30, 47, 39), (84, 13, 45), (38, 79, 51), (92, 45, 57),
   (46, 11, 63), (0, 77, 69), (54, 43, 75), (8, 9, 81), (62,
    75, 87), (16, 41, 93), (70, 7, 99)]
```

1.1.5 Constraints

For each player's standing and jump: `0 <= standing <= 99` and `1 <= jump <= 99`.

1.1.6 Testing

In order to test your function, type the following command on the terminal:

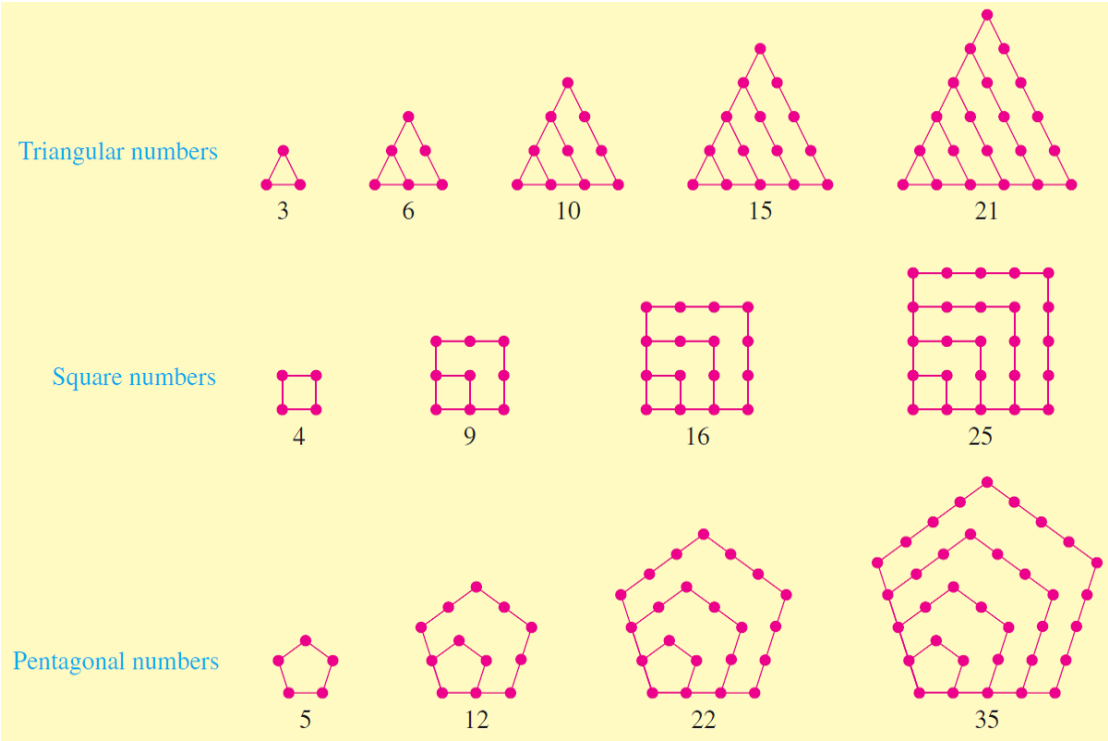
```
pytest tests/test_q1.py
```

1.2 Who wants to be a multipillionaire?

1.2.1 Background

On a particular game show, contestants are given a large supply of marbles and asked to create a shape of their one of the following shapes: triangle, square, pentagon. In round 1, each contestant chooses their shape and uses their marbles to make it. In each subsequent round, they grow their shape by adding a new layer of marbles to it. A goal number is defined. At the end of each round, the number of marbles in each player's shape is counted. If this number is divisible by the goal, the player is declared the winner. If no winner is found in a round, the game proceeds to the next round, until a winner is declared.

The diagram below shows the growth of some shapes and the total number of marbles at each stage.



1.2.2 Example

Anjum and Beenish are competing one night and the goal is 20. Anjum starts with a triangle (3 marbles). The number of marbles in her shape at the end of each round will be: 3, 6, 10, 15, 21, 28, 36, 45, 55, 66, 78, 91, 105, 120, ...

Beenish starts with a square (4 marbles). The number of marbles in her shape at the end of each round will be: 4, 9, 16, 25, 36, 49, 64, 81, 100, ...

At the end of Round 1, Anum has 3 marbles and Beenish has 4. None are divisible by the goal which is 20, so the game proceeds to the next round. At the end of the next round, the numbers of marbles are 6 and 9, and the game proceeds. The match occurs at the end of round 9 when Beenish has 100 marbles. 100 is divisible by 20 so the game ends and Beenish is declared the winner.

1.2.3 Challenge

Write a function named `multipillionaire` that accepts two arguments, sides of type list and goal of type int. sides contains the number of marbles that each player is starting with. The function returns a tuple containing the player number that wins and the total number of rounds played.

A tuple is a container like a list but uses parentheses, i.e. (and), as delimiters instead of the square brackets, i.e. [and], that a list uses. For example, (1, 2) is a tuple and [1, 2] is a list. Like a str, a tuple is immutable. Once created, it cannot be modified.

1.2.4 Note

If side is the number of sides in a player's shape, then in each round, the number of marbles that the player adds to their shape is: round * (sides - 2) + 1. For example, if side is 5, then in round 2 the player adds 2 * (5 - 2) + 1 = 7 new marbles and has a total of 5 + 7 = 12 marbles at the end of round 2. In the next round, she adds 3 * (5 - 2) + 1 = 10 new marbles and has a total of 12 + 10 = 22 marbles at the end of round 3.

1.2.5 Sample interaction

```
>>> multipillionaire([3, 4], 20)
(2, 9)

>>> multipillionaire([3, 4], 120)
(1, 14)

>>> multipillionaire([5, 4, 3], 17)
(1, 5)
```

1.2.6 Sample Interaction Explaination

In the first case, there are 2 players starting with 3 and 4 marbles respectively in round 1 and the goal is to reach a multiple of 20. The earliest that the goal is met is at the end of Round 9 when Player 2 reaches 100 marbles.

In the second case, there are 2 players starting with 3 and 4 marbles respectively in Round 1 and the goal is to reach a multiple of 120. The earliest that the goal is met is at the end of Round 14 when Player 1 reaches 120 marbles.

In the third case, there are 3 players starting with 5, 4, and 3 marbles respectively in Round 1 and the goal is to reach a multiple of 17. The earliest that the goal is met is at the end of Round 5 when Player 1 reaches 51 marbles.

1.2.7 Constraints

- `isinstance(sides, list) and isinstance(goal, int)`
- `all([isinstance(s, int) and 3 <= s <= 20 for s in sides])`
- `goal >= 2`

1.2.8 Testing

In order to test your function, type the following command on the terminal:

```
pytest tests/test_q2.py
```